

ECE 601 – Machine Learning for Engineers

Spring 2026 Course Project – Phase 4: Literature Search

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(a) Survey of State-of-the-Art Approaches

Skin lesion classification on the HAM10000 dataset has been an active research area in machine learning and medical image analysis. The following survey covers representative state-of-the-art approaches, organized by methodology.

1. EfficientNet-Based Approaches

Szegedy et al.’s EfficientNet family has become a popular backbone for skin lesion classification due to its strong accuracy-to-parameter-count ratio. Thomsen et al. [1] trained an EfficientNet-B1 model on the HAM10000 dataset using transfer learning and fine-tuning, achieving a test accuracy of **84.3%**. Per-class F1-scores were reported as follows: actinic keratoses (0.63), basal cell carcinoma (0.72), benign keratosis-like lesions (0.70), dermatofibroma (0.54), melanoma (0.58), and vascular lesions (0.80). The authors noted that class imbalance remained a significant challenge, particularly for minority classes such as dermatofibroma.

A more recent and powerful variant, EfficientNetV2-L with channel attention and a three-stage progressive learning strategy, was proposed by Karthik et al. [2]. This approach incorporated high-quality data balancing, large-scale augmentation, and explainable AI techniques (Grad-CAM and saliency maps) for diagnostic transparency. It achieved a total accuracy of **91.15%** and a macro F1-score of **85.45%** on HAM10000, representing a substantial improvement over simpler transfer learning baselines.

2. DenseNet-Based Approaches

DenseNet architectures, where every layer is connected to every other layer in a feed-forward manner, have demonstrated consistently strong performance on HAM10000. Thurnhofer-Hemsi et al. [3] evaluated 11 different CNN architectures on HAM10000 and found that DenseNet169 achieved the best performance at **92.2%** accuracy, **93.6%** sensitivity, and **93.3%** F1-score. A modified DenseNet201, in which the final classification head was replaced with global average pooling, five fully connected layers, dropout, and a softmax output, was shown to further improve upon DenseNet169, gaining 13.8% additional accuracy on HAM10000 [4]. These results suggest that dense connectivity and aggressive feature reuse are particularly well-suited to the fine-grained visual differences between skin lesion classes.

3. Vision Transformer (ViT) and Hybrid Approaches

Transformer-based architectures have emerged as strong competitors to CNNs for medical image classification. Datta et al. [5] achieved an accuracy of **93.4%** on HAM10000 by combining high-performing CNNs with soft attention layers, along with a preprocessing strategy that removed duplicate entries and applied both oversampling and undersampling for class balancing. The soft attention mechanism allowed the model to focus on diagnostically relevant regions of each lesion image.

A more recent study combining Vision Transformers (ViT) with EfficientNet in an ensemble with majority voting achieved **95.05%** accuracy on the ISIC 2018 dataset (which shares the same 7-class taxonomy as HAM10000), demonstrating that multi-scale attention combined with ensemble learning can push performance to near-clinical-grade accuracy [6]. Transformer-based models have been shown to outperform CNN-based approaches in several benchmarks, likely due to their ability to model long-range dependencies and global context in images.

4. Multimodal Learning Approaches

Beyond image-only classifiers, recent work has explored combining dermoscopic images with patient metadata (age, sex, lesion location) in multimodal deep learning frameworks. Adebiyi et al. [7] compared several deep learning models on HAM10000, including InceptionV3 (86.53% accuracy), DenseNet121, and an ALBEF multimodal model that jointly encodes images via a Vision Transformer and metadata text via BERT. The ALBEF model achieved the best accuracy of **94.11%** and an AUROC of **0.9426**, outperforming all image-only baselines. This demonstrates that patient metadata, while simple, provides complementary diagnostic signal that image features alone cannot capture.

5. Soft Attention with Imbalance-Aware Loss

Khoi et al. [8] proposed combining DenseNet, InceptionNet, and ResNet backbones with a soft attention mechanism that unsupervisedly extracts heat maps of the primary lesion region, combined with a novel imbalance-aware loss function. Using InceptionResNetV2 with soft attention, the method achieved **90%** accuracy, with mean precision, F1-score, recall, and AUC of 0.81, 0.81, 0.82, and 0.99 respectively on HAM10000. A lighter variant using MobileNetV3Large with soft attention achieved 86% accuracy while being 30 times faster at inference, demonstrating an attractive accuracy/efficiency trade-off for clinical deployment.

(b) Numerical Comparison with Literature

Table 1 presents a direct comparison of the performance achieved by our Phase 2 baseline (ResNet-18) and our best Phase 3 result (ResNet-50 with WeightedRandomSampler) against the state-of-the-art methods surveyed above.

Method	Source	Accuracy	Macro F1
ResNet-18 (our baseline, Phase 2)	This work	83.30%	0.754
ResNet-50 + WeightedSampler (Phase 3)	This work	85.12%	0.782
EfficientNet-B1	Thomsen et al., 2023	84.30%	—
InceptionResNetV2 + Soft Attention	Khoi et al., 2022	90.00%	0.81
EfficientNetV2-L + Channel Attention	Karthik et al., 2025	91.15%	0.855
DenseNet169	Thurnhofer et al., 2021	92.20%	0.933
CNN + Soft Attention (ensemble)	Datta et al., 2021	93.40%	—
ALBEF (Multimodal: Image + Metadata)	Adebiyi et al., 2024	94.11%	—
ViT + EfficientNet Ensemble	Vit Ensemble, 2024	95.05%	—

Table 1: Comparison of our approach against state-of-the-art methods on HAM10000. Dashes indicate metrics not reported in the original paper.

Our Phase 2 baseline (83.30% accuracy, macro F1 0.754) is competitive with simpler transfer learning baselines in the literature, such as the EfficientNet-B1 result of 84.30% from Thomsen et al. Our Phase 3 best result (85.12% accuracy, macro F1 0.782) slightly surpasses this. However, there is a clear performance gap between our approach and the state of the art. Methods that incorporate attention mechanisms, deeper or more modern backbones (DenseNet169, EfficientNetV2-L), ensemble strategies, or multimodal inputs consistently achieve accuracies in the 90–95% range. The primary sources of this gap appear to be: (1) our use of a relatively shallow backbone (ResNet-18/50) without attention, (2) the lack of patient metadata as an additional input modality, and (3) the absence of test-time augmentation or ensemble strategies.

(c) Available Implementations and Candidate for Phase 5

The following implementations are publicly available for the approaches surveyed above:

1. **Soft Attention + CNN (Khoi et al., 2022)** — Full code available on GitHub: <https://github.com/KhoiDOO/SLC-ID-SA>. Implements InceptionResNetV2 and MobileNetV3Large with soft attention and an imbalance-aware loss function. Achieves 90% accuracy on HAM10000.
2. **EfficientNet-B1 (Thomsen et al., 2023)** — Implementation available via the paper’s supplementary materials and reproducible using standard torchvision/timm libraries. Achieves 84.3% accuracy, closest to our own baseline in terms of methodology.
3. **DenseNet169 (Thurnhofer et al., 2021)** — Standard DenseNet architectures are available in torchvision. The specific configuration from the paper (DenseNet169 fine-tuned on HAM10000) can be reproduced directly using pretrained weights from torchvision and the same training setup described in the paper. Achieves 92.2% accuracy.

Proposed Candidate for Phase 5

We propose to implement and compare against the **Soft Attention + InceptionResNetV2 approach** from Khoi et al. (2022), available at:

<https://github.com/KhoiDOO/SLC-ID-SA>

This candidate is selected for the following reasons:

- The full code is publicly available and targets HAM10000 directly, making replication straightforward.
- The soft attention mechanism is conceptually grounded in course material (attention as a form of learned weighting) and provides a meaningful architectural contrast to our plain ResNet baseline.
- The imbalance-aware loss function is a direct extension of the weighted cross-entropy used in our own approach, allowing for a clean ablation of the attention mechanism's contribution.
- At 90% accuracy and macro F1 of 0.81, it represents a meaningful performance gap from our method (85.12%, 0.782) while remaining computationally feasible on an NVIDIA RTX 3060.
- As an alternative, DenseNet169 is also feasible via torchvision if the instructor prefers a simpler implementation with a larger performance gap.

References

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- [8] Khoi, D. et al. *Skin Lesion Classification on Imbalanced Data Using Deep Learning with Soft Attention*. Sensors (MDPI), 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9572097/>